Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Shaw, Benjamin Ph.D.		
Project Title:	Combustion of HAN-Based Monopropellant Droplets in Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Davis	**State:**	CA
Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/04/1999	**End Date:**	01/03/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NCC3-683		
Performance Goal No.:			
Performance Goal Text:			

These experiments involve burning unsupported and supported droplets within a pressure chamber. Droplets and flames are photographed with on-board video cameras (CCD and ICCD) to allow direct observation of liquid and gas behaviors. Data to be obtained will include transient droplet and gas-phase flame diameters, the occurrence of liquid bubbling inside droplets, the low- pressure deflagration limit, influences of fuel species on liquid consumption rates, radiant heat emissions, and gas thermal profiles.
Theoretical studies focus on extending previously developed analytical models of HAN-based liquid propellants to include phenomena such as non-equilibrium vaporization, bubbling, alternative chemical-kinetic schemes, gas-phase chemistry, and reactions of fuel species such as TEAN or methanol. The objective of this research is to study the combustion of monopropellant droplets and monopropellant droplet components in reduced-gravity environments so that spherical symmetry is strongly promoted. Hydroxylammonium nitrate (HAN) is a major constituent in a class of liquid monopropellants with many attractive characteristics and which display phenomena that differ from other liquid

Task Description:	monopropellants. The research will involve studying combustion of droplets containing HAN (which is oxygen rich), water, and fuel species such as TEAN, glycine and methanol. Droplets/particles initially in the mm size range are studied at pressures up to 30 atm. The experimental studies utilize reduced gravity to strongly promote spherical symmetry. This will allow observations of burning rates and flame structures to be made without the complicating effects of buoyant and forced convection. Normal gravity comarison experiments are also performed. The theoretical studies will complement the experimental work. HAN-based liquid monopropellant combustion is relevant to applications such as spacecraft thrusters and liquid-propellant guns. Fundamentally, there are processes associated with HAN-based liquid monopropellant combustion that are not well understood, e.g. laminar liquid surface deflagration rates, TEAN droplet combustion mechanisms and rates, and gas-phase flame extinction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 04/29/2010)